

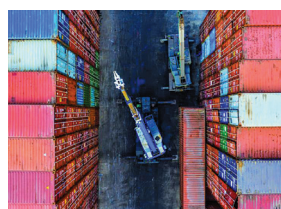
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

End-to-end asset tracking service

U-blox has introduced CloudTrack™, an end-to-end asset tracking service designed for the Internet of Things (IoT) sector. The service offers low-power



positioning, connectivity worldwide, and cloud integration. A key feature is its ability to

achieve six times the energy efficiency of traditional GNSS systems starting cold and transmitting data online. The service combines data from GNSS, cellular, and Wi-Fi sources for positioning, even in environments with poor GNSS signals or indoors—areas typically prone to rapid battery depletion. It offers a simplified experience by consolidating multiple location and connectivity needs under one provider. This approach eliminates the hassle of managing multiple contracts and vendors, supported by a straightforward pricing model.

U-blox

<https://www.u-blox.com>

Power modules to boost solar inverters' power

Onsemi has launched silicon and silicon carbide hybrid power integrated modules (PIMs) in an F5BP package, designed for enhancing the power output of utility-scale solar string inverters or energy storage system (ESS) applications. The modules increase power density and efficiency within the same footprint, raising the total system power of a solar inverter from 300kW to 350kW. This enhancement allows a one-gigawatt (GW) capacity utility-scale solar farm using these

modules to save nearly two megawatts (MW) per hour, equivalent to powering over 700 homes annually. These



modules also require fewer units to achieve the same power level as previous generations, reducing power device compo-

nent costs by more than 25%. The FS7 IGBTs reduce turn-off losses and lower switching losses by up to 8%, while the EliteSiC diodes enhance switching performance and decrease voltage flicker by 15% compared to previous models.

Onsemi

<https://www.onsemi.com>

Memory module IC for enhanced data processing

ABLIC has released the S-34HTS08AB SPD IC, designed for DDR5 memory modules. The module adds a HUB



function to facilitate device communication and supports the high-speed I3C standard—an evolution of the I2C

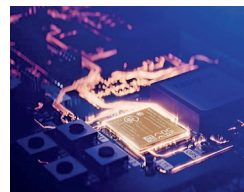
protocol—alongside traditional EEPROM and temperature sensor capabilities. It advances data processing capacities, making the IC suitable for both client and server DDR5 DIMMs. The company claims that the input-output circuits ensure high-speed, stable communication and broad compatibility with devices from other manufacturers, offering flexibility in component purchasing.

ABLIC

<https://www.ablic.com>

SiP for app development or use as a modem

Nordic Semiconductor has launched its system-in-package (SiP), the nRF9151, along with its development kit. The nRF9151 is an integrated, pre-certified



SiP that includes an MCU for application development or use as a cellular modem.

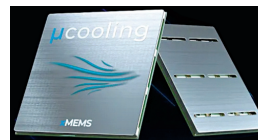
The nRF9151 provides a solution in cellular IoT technology. This package offers hardware with a programmable application processor, software, cloud services, development tools, and support. Supported by Nordic's SDK and cloud services, it also manages regulatory certifications and carrier pre-approvals, ensuring support for long-life IoT applications.

Nordic Semiconductor

<https://www.nordicsemi.com>

1mm-thin active micro-cooling 'fan on a chip'

xMEMS Labs has launched the xMEMS XMC-2400 µCooling™ chip. This is the first all-silicon, active micro-cooling



fan designed for mobile devices and artificial intelligence applications.

The chip introduces active, fan-based micro-cooling at the chip level, allowing manufacturers to incorporate cooling into smartphones, tablets, and other mobile devices. The chip measures only 1 millimetre in thickness. It has dimensions of 9.26 x 7.6 x 1.08 millimetres and